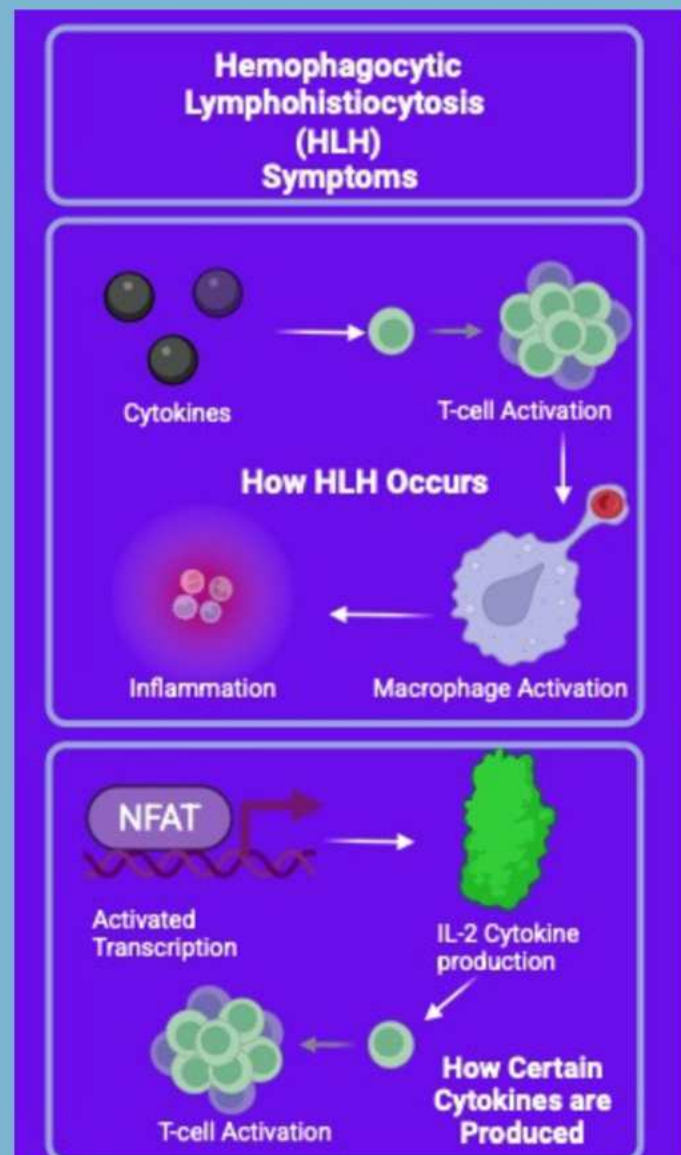


Introduction

Hemophagocytic Lymphohistiocytosis (HLH) is a rare **inflammatory syndrome** characterized by an overactive **immune system**. HLH has two forms. One result from **genetic mutations** affecting cytotoxic granule exocytosis, including **Perforin** (PRF), which is encoded by the Perforin-1 (PRF1) gene (Takushi et al., 2020). Studies have shown that PRF **regulates** the production of **cytokines** such as **IL-2**, where PRF **deficiency** increases cytotoxicity and can result in a **cytokine storm** (Trapani et al., 2015). The **other** has **no identifiable mutation** and is linked to **viruses** and **autoimmune disorders** (Keenan et al., 2024). Both forms, however, have **mortality rates near 30%** (Abdelhay et al., 2023). Current treatments use cyclosporine A, etoposide, and corticosteroids to treat HLH.

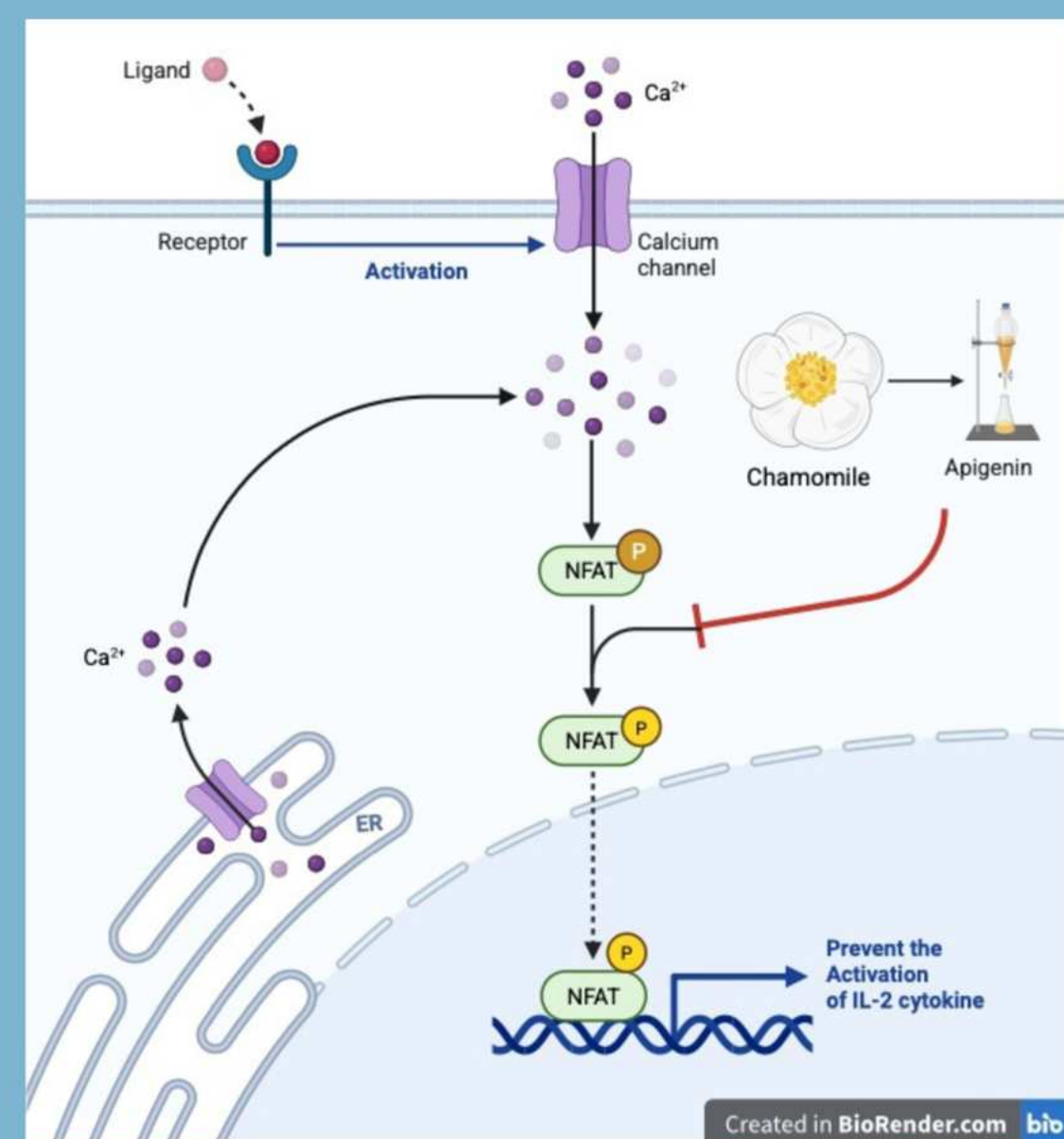
Figure 1: Visualization of HLH Symptoms and how it occurs



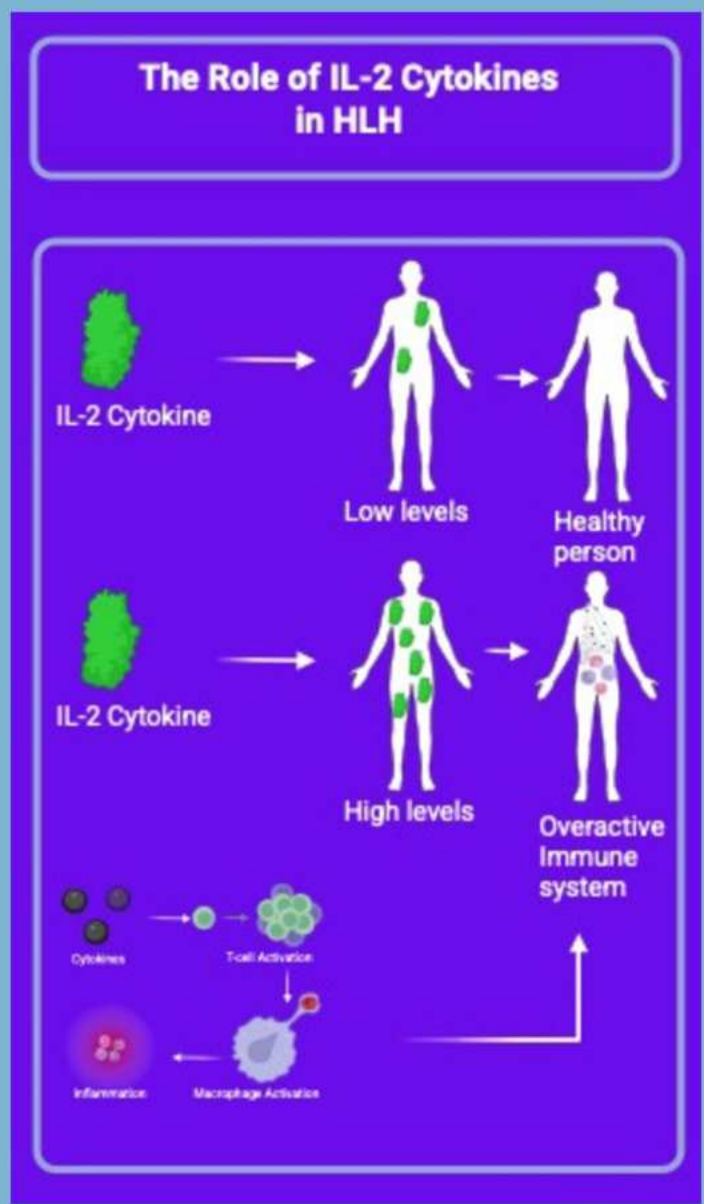
Note: Created in <https://BioRender.com>

Current research suggests that chamomile essential oil shows significant **anti-inflammatory properties** by reducing CD4+ **T cell activation**, which is involved in the inflammatory processes (De Cicco et al., 2023). **Chamomile extract** contains flavonoids like **apigenin** with antioxidant and **anti-inflammatory properties** (Drif et al., 2024). In addition, studies have also shown that the **dephosphorylated** nuclear factor of activated T-cells (NFAT) has **strong binding** to the PRF1 gene promoter region. Previous research has shown that **inhibition** of the nuclear translocation of **NFAT** reduces the expression of **IL-2** (Lee et al., 2018).

Figure 2: Visualization of how Apigenin affects IL-2



Note: Created in <https://BioRender.com>



Note: Created in <https://BioRender.com>

Purpose & Novelty

The purpose of the study was to see if using different **concentrations** of a **natural compound**, such as **apigenin**, would result in **varying capabilities** of **reduced inflammation** as seen by the **IL-2 cytokine**. It is known that **IL-2 promotes T-cell proliferation** and has been shown to induce **inflammatory responses** (Ross & Cantrell, 2018). It is also known that **dephosphorylated NFAT** allows for IL-2 to be expressed and affects T-cell proliferation (Lee et al., 2018). The novelty of the study is due to the **limitations and gaps** in HLH research to use **natural targeted treatments** to **inhibit NFAT** dephosphorylation to inhibit IL-2 production *in vitro*.

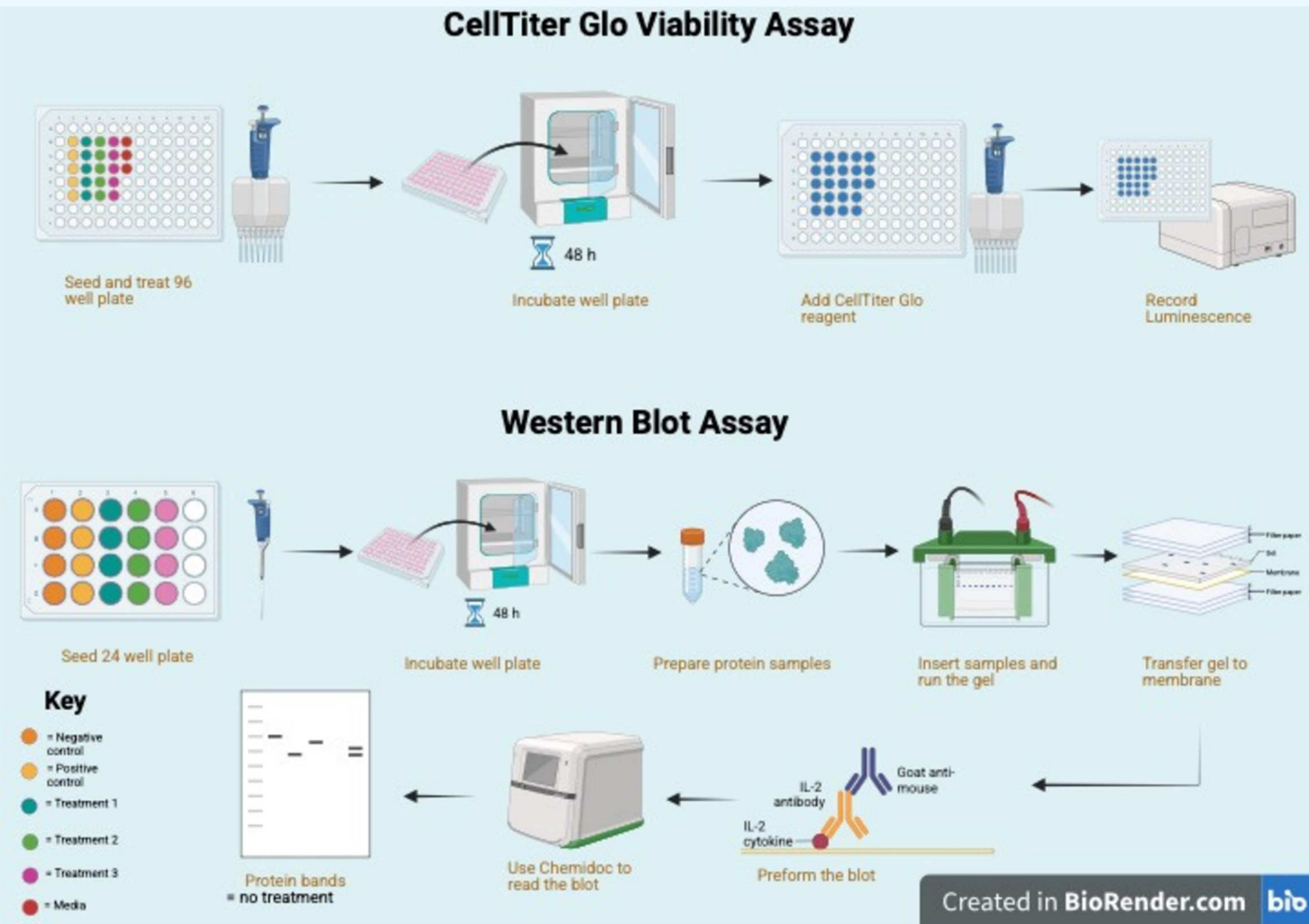
To investigate this study, it was hypothesized that if the different concentrations of apigenin used **prevents T cell activation** through IL-2, then the **amount** of IL-2 cytokines **produced** by the Jurkat E6.1 cells should **decrease** because the dephosphorylated can not **activate IL-2** production.

The Effect of Apigenin on the Production of IL-2 Via NFAT Signaling

Cell

Methodology

Figure 2: Experimental Procedure



Constants

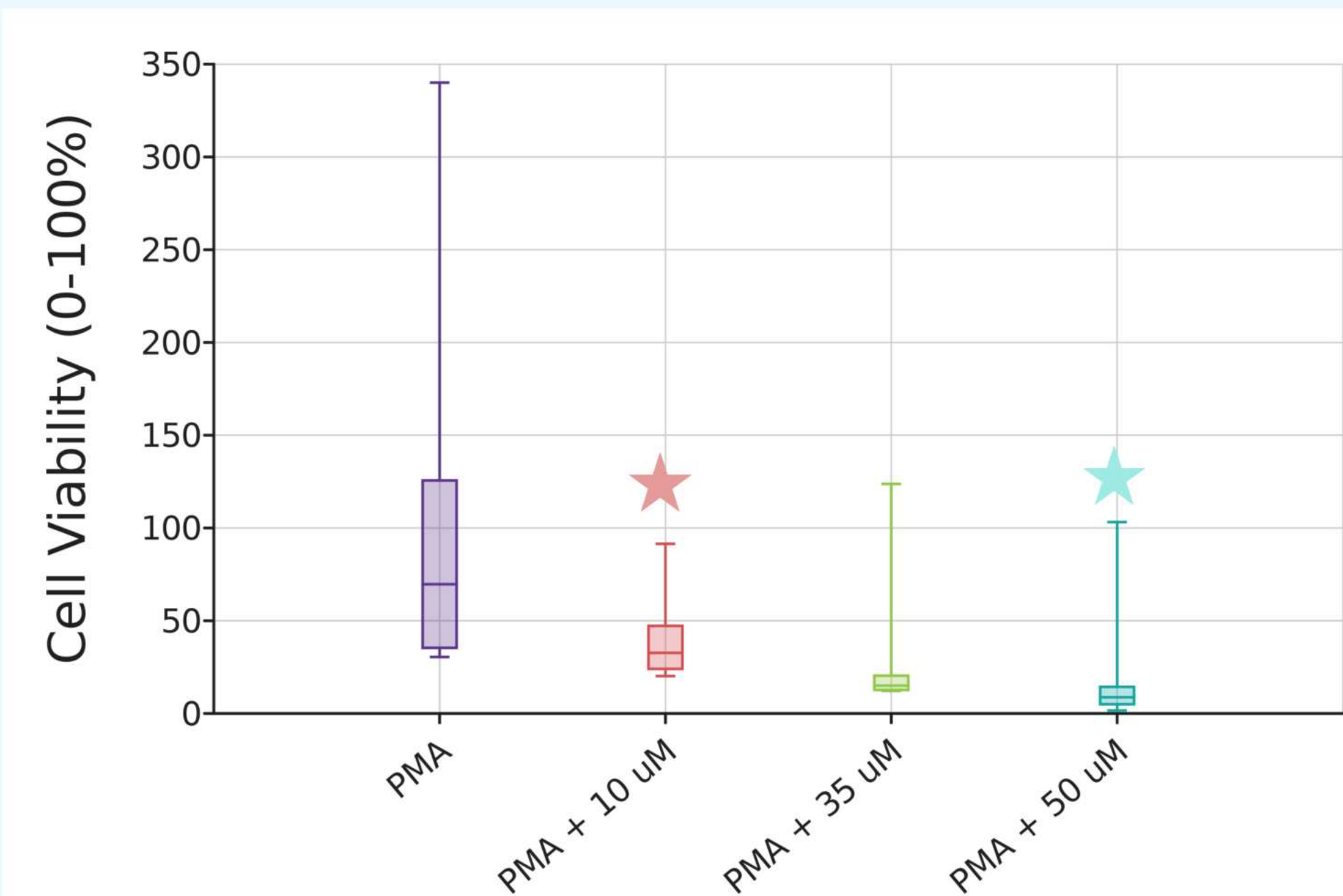
- Jurkat E6.1 cells
- Growth media (90% RPMI 1640 + FBS)
- Storage temperature (37 C CO2 incubator)
- PMA + Ionomycin concentration (10X solution)
- CellTiter Glo reagent volume (100 uL)

Variables

- Independent variable: Apigenin of different concentrations of 10uM, 35uM, 50uM
- Dependent Variable: Amount of IL-2 protein measured using a western blot.
- Negative control: Cells with growth media.
- Positive control: Cells are grown with just PMA + Ionomycin

Data & Results

Combined Cell Viability for Different Concentration of Apigenin

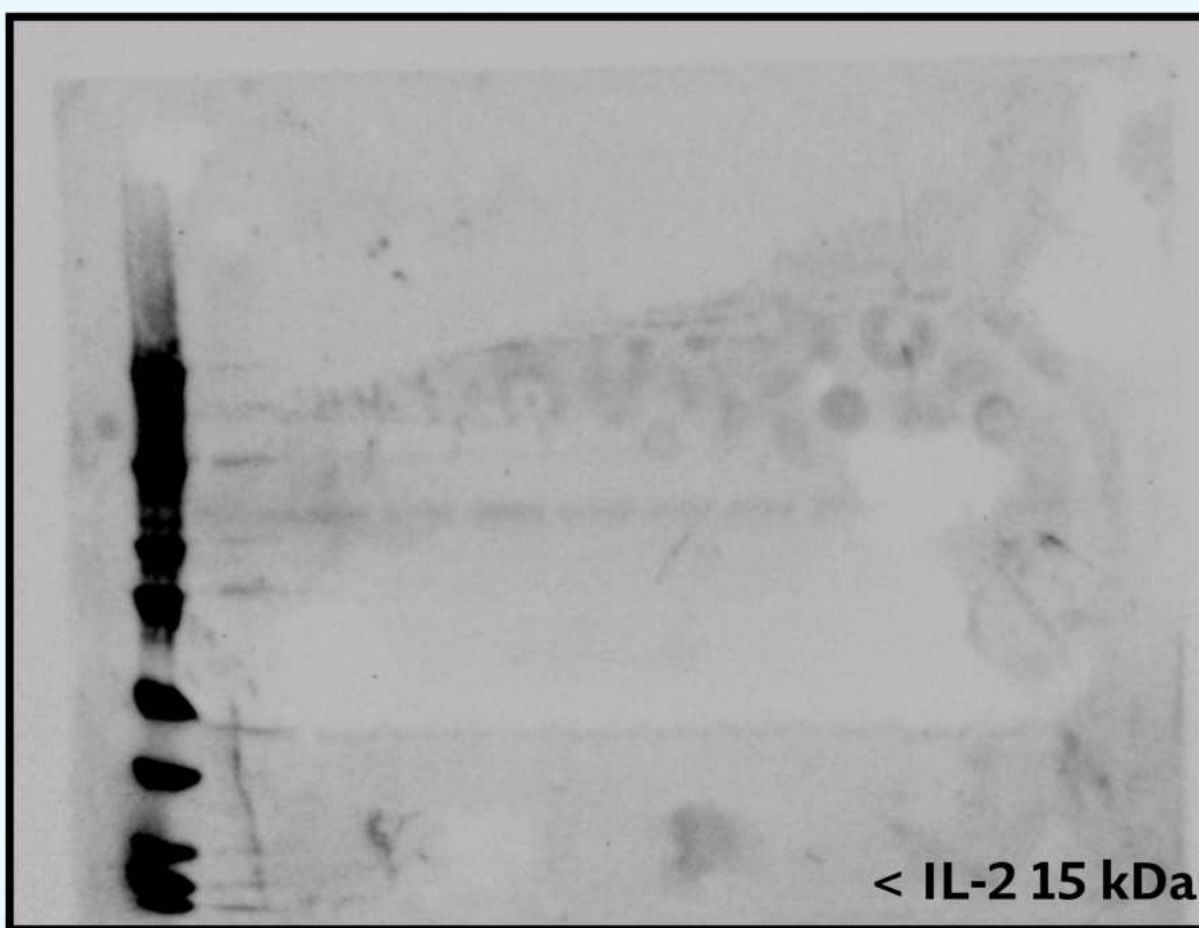


Treatment (PMA + Apigenin concentration)

Note: Box and whiskers plot shows the median value (line), interquartile range (box), and the extent of the data (whiskers above and below at min / max data points).

Note: Created in Data Classroom. Shows the relationship between the concentration and cell viability Key: PMA = Cell Stimulation Cocktail containing PMA and Ionomycin, PMA + 10 uM = Cell Stimulation Cocktail containing PMA and Ionomycin treated with 10 micromolar of Apigenin, PMA + 35 uM = Cell Stimulation Cocktail containing PMA and Ionomycin treated with 35 micromolar of Apigenin, PMA + 50 uM = Cell Stimulation Cocktail containing PMA and Ionomycin treated with 50 micromolar of Apigenin, RLU = relative light units. Note, PMA = positive control as it is known to stimulate cell activity, therefore increasing ATP levels in cells.

Trial 1: Western Blot for IL-2 Using IL-2 Monoclonal Antibody



Trial 2: Western Blot for IL-2 Using IL-2 Monoclonal Antibody

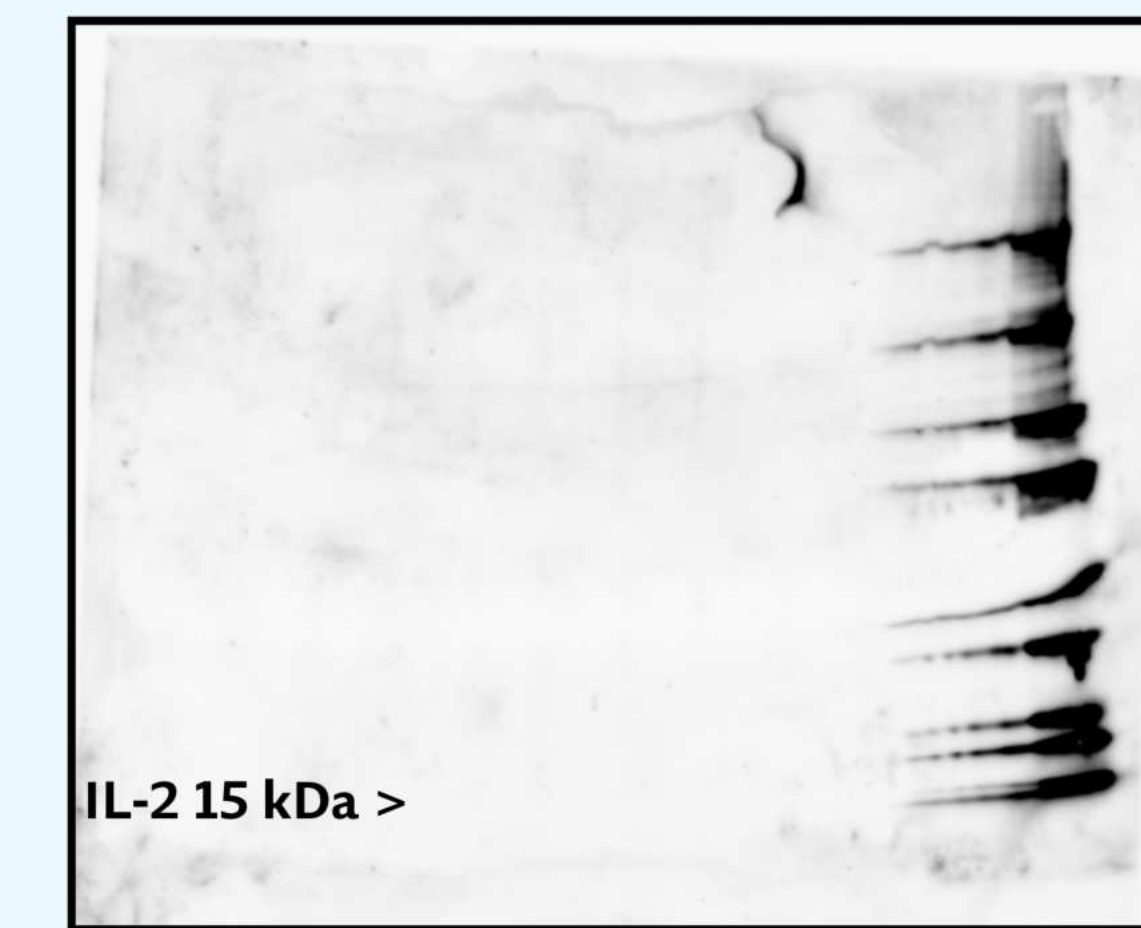


Table 1: Post Hoc Dunn's Test CellTiter Glo Statistical Analysis

PMA - PMA + 10 uM	★	1	Means: that the groups are NOT different
PMA - PMA + 35 uM		0.01	Means: very strong confidence that the groups are different
PMA - PMA + 50 uM	★	<0.01	Means: extremely strong confidence that the groups are different
PMA + 10 uM - PMA + 35 uM		0.47	Means: little to no indication that the groups might be different
PMA + 10 uM - PMA + 50 uM	★	0.04	Means: strong confidence that the groups are different
PMA + 35 uM - PMA + 50 uM	★	1	Means: that the groups are NOT different

All images, unless indicated otherwise, have been created by the researcher.

Evaluation

Based on the **Kruskal-Wallis** test performed, the conclusion is that there **is a significant difference** between different concentrations of apigenin and the resulting cell viability. Since the **p-value is <0.01**, that means that there is an **extremely strong confidence** that there is a significant difference between the different concentrations and the resulting cell viability. Based on the data collected, the **difference** between **no apigenin** and the **highest apigenin** concentration shows that there is a significant decrease in cell viability of the combined processed data, which shows that the median is close to **75% with no apigenin** and to close to **10%** at the **highest apigenin** concentration. This shows a decrease in median cell viability of about **65%**. Since the conclusion is that there is a significant difference, this means that the data and the statistics shows that the **cell viability decreases** as the concentration of **apigenin increases**. This suggests that the apigenin **concentrations** that are greater than or equal to **50uM** are more **harmful** to cell viability than any **potential benefits** that may be observed at that concentration. Another possible result is that concentrations that are greater than or equal to 50uM could result in a significant **decrease in cellular activity** due to the **potency** of the treatment rather than the decrease in the **overall viability** of the cells. The CellTiter Glo viability assay uses the **ATP** produced by the cell to measure **cell viability**, meaning that the significant decrease may **not** be due to cell **death**, but rather due to the decrease in ATP produced by the cell. This could suggest that apigenin can be used in **HLH treatment**, should there be a significant difference in IL-2 expression as it could **reduce the immune cell activity** and **not cause excessive toxicity** to healthy cells.

Following up on the Kruskal-Wallis, the **post-hoc Dunn's test** was performed to determine the **significance** of the cell viability **between** the **different groups** of apigenin concentrations. Based on Table 1 the statistical analysis is **consistent** with the difference in the median between these groups mentioned previously, and provides greater **indication** that apigenin concentrations **greater** than or **equal to 50uM** are more harmful compared to potential benefits. In addition, between the **p-value of 0.04** and the **p-value of 1**, between the positive control and 10uM group, the Dunn's test indicates that apigenin concentrations that are **less** than or **equal to 10uM** have potential benefits that are greater than any potential harm. As there was little to no significance between the 10uM and 35uM treatment groups, these **indicate** that it is likely that apigenin concentrations **greater than 35uM** are likely to have a **decrease on the return of value**.

Future Work

As the western blot was hard to read, in the **future**, I would proceed by trying to **quantify** my **IL-2 expression** using an **ELISA assay**, which is ideal to measure **small proteins** that are not largely expressed. However, based on the statistical conclusion from this research, I would **adjust** my treatment **concentrations** to be between 10uM and 35uM. Should apigenin show that it can reduce IL-2 expression, I would **continue** my project by researching the **minimum and maximum** concentrations of apigenin that affect the amount of IL-2 expression.

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